

Data-driven complementary policy development for photovoltaic industry facing resource constraints: A case study of Zhejiang Province in China

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ABSTRACT

Accelerating renewable energy development is critical for climate goals, with photovoltaic (PV) power emerging as a leading technology. However, PV's land-intensive nature triggers a “green-green dilemma” between climate targets and natural resource conservation, that is a globally prevalent challenge particularly acute in Zhejiang Province, China, where land constraints hamper large-scale PV expansion. To address this issue, this study proposes six PV-industry complementary modes including mountain-based agrophotovoltaics and offshore wind-solar hybrids, etc. Furthermore, we introduce the data-driven complementary development (DCD) framework, an innovative approach integrating K-means clustering, Mann–Whitney *U* test, and significance-based tier-searching (STS) algorithm to quantify subregional development potential for each mode. Unlike traditional clustering methods, the DCD framework enhances statistical robustness in potential identification. Applied to 90 subregions in Zhejiang, this framework reveals that most areas are suited for at least one complementary mode, providing actionable insights for local policymakers. This study makes three key contributions: enriching PV complementary development literature, proposing a replicable decision-support tool for resource-constrained regions, and offering targeted strategies to balance renewable energy growth and natural resource preservation in Zhejiang.

1. Introduction

Accelerating the development of renewable energy sources is widely acknowledged as a critical strategy for achieving climate goals [1]. RESs have garnered increasing attention in distribution systems and are anticipated to play a critical role in the energy landscape in the coming years. Photovoltaic (PV) energy has emerged as the leading RES technology and is expected to contribute more than 4500 GW of electricity by 2050 to meet global electricity demand. The utilization of solar energy plants is widely recognized as an exceptionally advantageous method for achieving climatic goals [2].

However, a green–green dilemma arises because of the contradiction between conserving natural resources (such as land) and achieving

climate goals. Compared with other renewables such as biomass energy plants and wind farms, solar energy plants are inherently land intensive in terms of the power produced per unit area [3]. Consequently, the expansion of solar energy consumes vast amounts of land resources, adversely impacting natural resource preservation, including deforestation, grassland reduction, and the depletion of forested areas in mountainous regions [4,5]; [6]. Furthermore, the impact of solar energy on land resources extends beyond its immediate physical footprint when new plants are constructed in natural ecosystems or former wilderness areas. In regions with scarce land resources, uncontrolled growth of solar energy facilities can lead to bird mortality, loss of biodiversity and habitat, and the presence of hazardous chemicals in solar panels (Dhar et al., 2020; [4,5]). This contradiction may lead to

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suboptimal outcomes of PV initiatives [7]. Therefore, there is a pressing need to align developing “affordable and clean energy” (Goal 7) with preventing the degradation of “life on land” (Goal 15) (UN, 2015).

This dilemma, which is a global challenge that requires urgent attention, is particularly pronounced in Zhejiang Province, China. As a highly developed coastal province in eastern China, Zhejiang consumed a staggering 619.2 billion kWh of electricity in 2023, with a significant 33.8% imported from other provinces. In response, Zhejiang has unveiled the ‘14th Five-Year Plan for Energy Development in Zhejiang Province’ aimed at reducing external energy dependence. This plan provides policy incentives to spur the rapid growth of renewable energy. Nevertheless, the construction of renewable energy facilities, particularly photovoltaic (PV) panels, has led to the encroachment upon and damage of vast swathes of valuable and prime land, including forestland. This poses a serious threat to breaking China’s ‘red line’ for land resource preservation. Consequently, the large-scale development of renewable energy, particularly photovoltaic power, is being hampered by land resource constraints. Zhejiang Province must strike a balance between vigorously pursuing renewable energy generation to meet the growing demand for electricity and preserving natural resources. However, how to optimize the development of renewable energy while respecting the constraints imposed by natural resources, particularly land resources, remains unclear. Zhejiang Province has not yet promulgated specific policies addressing this issue.

To address the aforementioned challenge on the basis of the unique characteristics of land resources in Zhejiang, we propose six PV-based complementary modes that seamlessly integrate PV generation with other industries, such as agriculture and animal husbandry. These modes include mountain-based agrophotovoltaics (MAPV), flatland-based agrophotovoltaics (FAPV), fishery-complementary photovoltaics (FPV), pasture-based photovoltaics (PPV), offshore wind-solar hybrid (OWS), and mountain-based wind-solar hybrid (MWS). Furthermore, to accurately quantify the development potential of each complementary mode in various subregions of Zhejiang, we introduce a comprehensive strategy formulation framework named data-driven complementary development (DCD). This framework aims to promote the sustainable development of solar energy by efficiently utilizing limited natural resources. The DCD framework involves integrating renewable energy production facilities, particularly PV in this context, with existing primary industries to transcend traditional development patterns.

To assess the suitability of the six complementary solar power development modes in Zhejiang, we followed a structured procedure within the DCD framework. Initially, we designated each county or district as a subregion. We subsequently gathered comprehensive data on sunlight intensity, wind speed, and land resources (including mountainous and aquatic areas) for each subregion and performed data cleaning and analysis to establish a foundational understanding of their resource endowments. For each complementary mode, we applied K-means clustering to the dataset to categorize Zhejiang’s 90 subregions into distinct groups. We then utilized the Mann–Whitney *U* test to statistically validate the significance of differences in key features between groups for each complementary mode. Ultimately, we applied the proposed significance-based tier-searching (STS) algorithm to pinpoint subregion groups offering the greatest potential for each complementary mode.

By adopting the DCD framework, we can accurately gauge the potential of subregions for various complementary modes and articulate the implementation strategies for each. This, in turn, aids in the formulation of targeted industrial policies by local governments, ensuring efficient and sustainable solar energy development tailored to Zhejiang’s unique geographical and resource conditions. In accordance with the results of the potential analysis on real-world data, some subregions in Zhejiang Province have successfully integrated PV energy with local industry.

The contributions of this study are threefold. First, we enrich the literature and practices related to renewable energy by introducing six

PV-based complementary modes tailored to the development of the PV industry, considering the unique endowment of natural resources. This framework is highly practical for developing complementary strategies within resource-limited contexts.

Second, we propose a data-driven framework to identify the potential of a subregion with respect to different complementary modes balancing natural resource preservation and energy industry development. In contrast to traditional clustering-based methods [8], the DCD framework achieves statistically more robust potential identification through the integration of K-means clustering, the Mann–Whitney *U* test, and the proposed STS algorithm. Moreover, this framework is a replicable decision-support tool for other resource-constrained contexts facing similar trade-offs.

Finally, we provide strategic insights for promoting the renewable energy industry in Zhejiang, China. By applying the DCD framework to real-world data, we identify five complementary modes for prioritizing PV industry development in the province. Our analysis reveals that most subregions are suited for at least one mode, although a few remain unsuitable for any. These findings offer actionable guidance for policy-makers and stakeholders in Zhejiang to pursue sustainable and efficient renewable energy development and investment strategies.

The remainder of this paper is structured as follows: Section 2 introduces the methodologies of the DCD framework. In Section 3, we present the data collection and visualization analysis and enumerate the empirical results obtained by applying the DCD framework in the context of Zhejiang. Section 4 introduces a discussion in relation to the literature and policy insights. Finally, we conclude the study in Section 5.

2. Methodology

The subregions in Zhejiang Province exhibit significantly different resource endowments, leading to varying possibilities for developing different PV complementing modes. Hence, it is necessary to classify and examine more than 90 subregions. In this study, we propose a DCD framework to identify the potential of different complementary modes for each subregion.

The DCD framework involves four key steps: complementary mode development, K-means clustering analysis, significance tests, and the identification of potential regional complementary modes. The development of complementary modes is detailed in Section 2. K-means clustering analysis has been widely used to address data clustering problems [9]. It remains a popular data-driven analysis method because of its high efficiency and concise algorithm [10]. Given that the data for each resource endowment are typically represented by numerical indices, which can be used to calculate the Euclidean distance metric in vector space, the K-means clustering algorithm is considered an effective method for categorizing subregions into distinct clusters on the basis of the corresponding indices. This algorithm is chosen because it allows us to further determine whether or not the differences between these clusters are significant. We employ the Mann–Whitney *U* test, a nonparametric method, to compare the differences between the samples of two independent clusters because the sample sizes are small and the distributions are not necessarily normal. The process of the DCD framework is illustrated in Fig. 1.

2.1. PV-based complementary mode development

The complementary development of associated industries and renewable energies has been the focus of PV-based complementary development schemes. For instance, Goetzberger and Zastrow [11] emphasized the coexistence of solar radiation conversion and plant cultivation. Recently, many researchers have analysed the optimization of the cooperative relationship between PV and agriculture (Fig. 2(a)). This includes the selection of suitable shade-tolerant crops that can be planted below PV panels [12] and the application of solar radiation

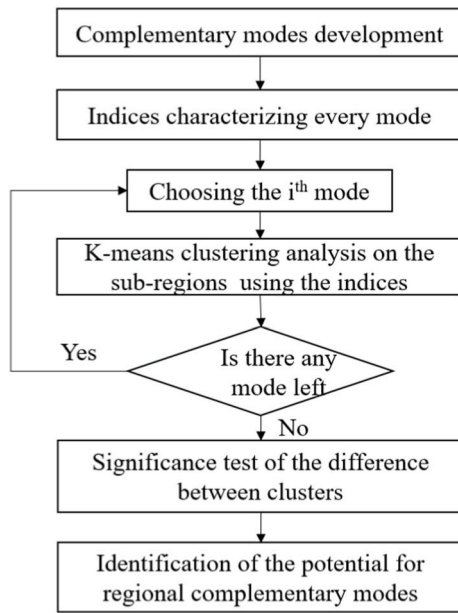


Fig. 1. Flow chart of the data-driven complementary development framework.

tracking systems to increase total crop yield [13]. On the basis of the diversity and complexity of the terrain in Zhejiang Province, as well as the crop species and natural conditions that restrict crop growth, this study explores two complements.

The breeding industry, mainly fisheries and livestock, can be integrated with PV systems. PV panels can be installed on shed roofs, livestock enclosures, and fish farms. For instance, Li et al. [14] reported that lower water body surface temperatures result in decreased PV panel temperatures and enhanced PV conversion efficiency. Furthermore, the interconnected symbiotic system involving rabbit husbandry and PV power generation has numerous mutually benefits [15,16]. Fig. 2(b, c) illustrates two types of integrated PV breeding: FPV and PPV.

Moreover, factors such as season, topography, and climate may

affect the stability and volatility of solar and wind energy. The exploitation of two complementary resources has given rise to a new type of energy generation system, such as OWS systems [17]. As complementary systems, PV and wind have relatively weak effects on the power grid, resulting in reduced grid disruptions. Consequently, the grid is more willing to accept renewable energy [18]. Thus, the stability of the hybrid system improves the complementary economic efficiency and grid acceptance of renewable energy. Moreover, meteorological factors such as solar radiation and wind speed affect wind-solar interconnected generation system energy storage [17]. In this study, we categorize the wind-solar complementary systems of Zhejiang Province into coast-based and mountain-based systems according to the distribution of natural resources with high wind speeds.

In summary, three complementary strategies suitable for the development of the PV industry in Zhejiang Province have been identified: APV [12], PV for breeding (Zhang et al., 2022), and wind-solar hybrid (Li et al., 2013). Specifically, APV consists of two modes: MAPV and FAPV. PV for breeding can be categorized into two types: FPV [19] and PPV [20]. Wind-solar hybrid systems include OWSs [21] and MWSs [22]. All six types of complementary modes are shown in Table 1.

2.2. Data preparation

The data preparation comprises the following steps.

- (1) **Data Integration and Standardization:** Data are collected from different sources. We first create a unified data schema to harmonize the disparate datasets. All the numerical values are

Table 1

Six types of complementary modes for PV power.

Mode category	Mode
Agrophotovoltaics	Mountain-based agrophotovoltaics (MAPV)
	Flatland-based agrophotovoltaics (FAPV)
Photovoltaics for breeding	Fishery complementary photovoltaics (FPV)
	Pasture-based photovoltaics (PPV)
	Offshore-based wind-solar hybrid (OWS)
Wind-solar hybrid	Mountain-based wind-solar hybrid (MWS)

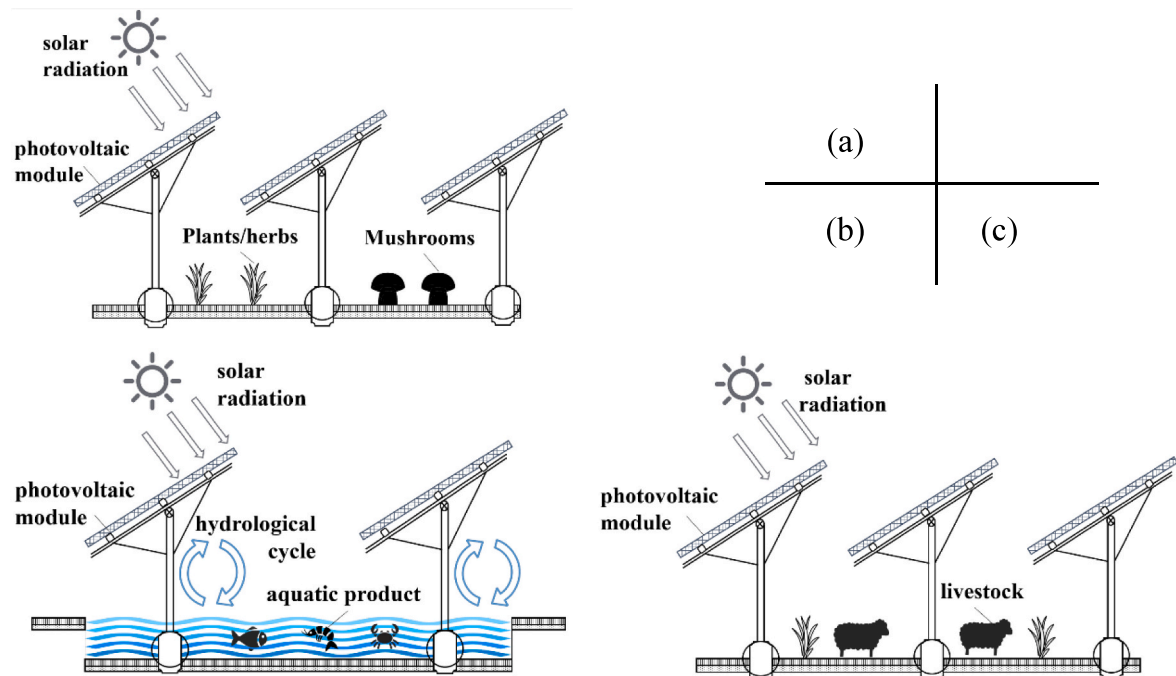


Fig. 2. Sketch map of (a) APV, (b) FPV and (c) PPV.

subsequently converted to a consistent unit of measurement to enable accurate comparison and analysis. Moreover, terms for technologies (such as “PV”, “photovoltaic”, “solar”) and areas of land resources are standardized into a single, consistent set of nomenclature across all records.

- (2) **Data Cleaning and Imputation:** Records with critical missing fields are identified and excluded from the analysis. For less critical fields with sporadic missing data, we employ median imputation for numerical data and mode imputation for categorical data to preserve the sample size where appropriate.
- (3) **Feature Engineering for Clustering:** Prior to applying the K-means clustering algorithm, all numerical input features are normalized using Z-score standardization.

2.3. K-means clustering analysis

We choose K-means clustering for two reasons: (1) K-means has a linear time complexity ($O(n)$), enabling highly efficient processing of datasets such as ours containing multiple regions and numerous variables. Compared with many hierarchical clustering methods, it demonstrates significant computational advantages when handling large sample sizes. (2) All the variables used in our study (including solar radiation intensity, wind speed, and land area) are continuous numerical variables. K-means employs Euclidean distance to measure similarity, which is particularly effective for the distribution patterns of such data, enabling natural grouping of regions on the basis of their numerical features.

This algorithm partitions the data into K clusters using a distance function, with the value of K specified in advance. Various criteria can be employed to determine the appropriate value of K [23]. In this study, the elbow method was used to identify the optimal K for each complementary mode. The K-means clustering procedure is as follows:

Input: Dataset $D = \{x_1, x_2, \dots, x_m\}$, the number of clusters K .
Process:
1. Randomly select K initial samples from D as the initial mean vectors $\{\mu_1, \mu_2, \dots, \mu_K\}$.
2. Repeat
3. Let $C_i = \emptyset$ ($1 \leq i \leq K$)
4. For $j = 1, 2, \dots, n$ do
5. Calculate the distance between sample x_j and each mean vector μ_i : $d_{ij} = \ x_j - \mu_i\ _2$
6. Determine the cluster label of x_j based on the nearest mean vector: $\hat{a}_j = \arg \min_{i \in \{1, 2, \dots, K\}} d_{ji}$
7. Assign the samples to their corresponding cluster $C_{\hat{a}_j} = C_{\hat{a}_j} \cup \{x_j\}$
9. End for
9. For $i = 1, 2, \dots, K$:
10. Calculate the new mean vector $\mu'_i = \frac{1}{ C_i } \sum_{x \in C_i} x$
11. If $\mu'_i = \mu_i$, then
12. Update the current mean vector μ_i to μ'_i .
13. Else
14. Keep the current mean vector unchanged
15. End if
16. End for
17. Until the current mean vectors remain unchanged.
Output: $\mathcal{C} = \{C_1, C_2, \dots, C_K\}$

2.4. Significance test of the difference between clusters

We chose the Mann–Whitney U test to statistically verify the significance of the difference between clusters because it is a nonparametric test used to determine whether two independent samples originate from the same distribution. Given that the K-means clustering analysis categorized 90 subregions into several clusters, we conducted significance tests using the outcomes of the K-means clustering. The procedure of the Mann–Whitney U test is as follows.

Input: Two independent samples: $S_1 = \{S_{1,1}, S_{1,2}, S_{1,3}, \dots, S_{1,n_1}\}; S_2 = \{S_{2,1}, S_{2,2}, S_{2,3}, \dots, S_{2,n_2}\}.$

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(continued)

Input: Two independent samples: $S_1 = \{S_{1,1}, S_{1,2}, S_{1,3}, \dots, S_{1,n_1}\}; S_2 = \{S_{2,1}, S_{2,2}, S_{2,3}, \dots, S_{2,n_2}\}.$
Process:
1. Ranking the Data Combine the two samples into one dataset and rank all the observations from smallest to largest. If there are ties (i.e., identical values) Then, assign the average rank to each tied value Denote the ranking of sample S_{ij} , denoted by R_i^j
2. Calculation of U Calculate the sum of the ranks of the observations in Sample set S_i : $R_i = \sum_j R_i^j$ ($i \in \{1, 2\}$) Calculate the U statistic: $U_i = n_1 \cdot n_2 + \frac{n_i(n_i + 1)}{2} - R_i$ The test statistic $U = \min \{U_1, U_2\}$
3. Significance identification Compare the generated U statistic with the critical values obtained from the Mann–Whitney U distribution tables.
Output: The value of the Mann–Whitney U test statistic and the associated significance level.

There are crucial values for the U statistic at different sample sizes and significance levels (usually $\alpha = 0.05$). If the estimated U value is less than or equal to the crucial value in these tables, the result is considered statistically significant. Consequently, the observed disparities in rank between the two groups are unlikely to have occurred by mere coincidence, resulting in the dismissal of the null hypothesis at the specified level of significance.

2.5. Identification of the potential for regional complementary modes

We proposed an STS algorithm that uses the results of the U test to identify the clusters with potential for the development of each complementary mode. Initially, we defined the ‘tier cluster’ as a cluster comprising many clusters (generated by the K-means algorithm) without a statistically significant difference. However, statistical analysis revealed a considerable difference in the features of one-tier clusters compared with those of adjacent tiers. For instance, the area of mountains are significantly larger in the first-tier cluster than in the second-tier cluster.

The STS algorithm consists of six steps: (1) we repeatedly choose one complementary mode, as shown in Loop #1 of Algorithm I. (2) With respect to the i^{th} mode, we rank the clusters for every index (see Loops #2 and #3); i.e., for the complementary mode MAPV, if we focus on the index of the area of the mountains, the clusters should be ranked as $\{C_1^1, C_2^1, \dots, C_{n_1}^1\}$. (3) For the j^{th} cluster in the i^{th} mode, we find the clusters in which the difference from cluster j is insignificant in the value of feature k_i^j . We define the number of clusters as τ_{ij}^k ($0 \leq \tau_{ij}^k \leq n_i - j$). (4) In the i^{th} mode, we choose the maximum τ_{ij}^k from those of all the indices. (5) We aggregate the subregions in cluster set $\{j+1, \dots, j+\tau_{ij}^k\}$ to form the l^{th} -tier cluster T_i^l (l is the number of tiers). (6) We iteratively repeat steps (1)–(5) until all six modes have been traversed. The procedure is illustrated in Algorithm I.

Algorithm I

Significance-based tier-searching algorithm

Input:
1. Clusters for each complementary mode, e.g. there are $n_i n_i$ clusters for the $i^{\text{th}} i^{\text{th}}$ mode.
2. For the $j^{\text{th}} j^{\text{th}}$ cluster $C_i^j C_i^j$ in the $i^{\text{th}} i^{\text{th}}$ complementary mode, there are $m_j m_j$ subregions, $i \in \{1, \dots, 6\} i \in \{1, \dots, 6\}, j \in \{1, \dots, n_i\} j \in \{1, \dots, n_i\}.$

(continued on next page)

Algorithm I (continued)

Input:

3. The index vector $\mathbf{V}_{k_i} \mathbf{V}_{k_i}^j$ for a subregion $k_i^j k_i^j$ in cluster $C_i^j C_i^j$, $\mathbf{V}_{k_i} = [v_1, \dots, v_{q_{ij}}] \mathbf{V}_{k_i}^j = [v_1, \dots, v_{q_{ij}}^j]$, $\mathbf{k}_i \in \{1, \dots, q_{ij}^k\} \mathbf{k}_i^j \in \{1, \dots, q_{ij}^k\}$.

Process:

Repeat

For $i = 1, 2, \dots, 6i = 1, 2, \dots, 6$ do/#1: Looping through six modes/

Repeat

$j = 1j = 1$

For $l = 1, 2, \dots, n_i l = 1, 2, \dots, n_i$ do/#2: Looping through $n_i n_i$ clusters in each mode/

Repeat

For $k_i = 1, 2, \dots, q_{ij}^k k_i = 1, 2, \dots, q_{ij}^k$ do/#3: Looping through q_{ij} indices in each cluster/

Rank the clusters according to the value of $k_i^j k_i^j$ and obtain the ranked cluster set $\{j, \dots, n_i\} \{j, \dots, n_i\}$.

Find the clusters in $\{j+1, \dots, n_i\} \{j+1, \dots, n_i\}$ showing that the difference with cluster jj is insignificant, the number of clusters is denoted as $\tau_{ij}^k, 0 \leq \tau_{ij}^k \leq n_i - j\tau_{ij}^k$, $0 \leq \tau_{ij}^k \leq n_i - j$.

End for

$\tau_{ij}^k = \max\{\tau_{ij}^k\} \tau_{ij}^k = \max\{\tau_{ij}^k\}$

Aggregate the subregions in cluster set $\{j+1, \dots, j+\tau_{ij}^k\} \{j+1, \dots, j+\tau_{ij}^k\}$ into the l^{th} tier cluster $T_i^l T_i^l$.

$j := j + \tau_{ij}^k j := j + \tau_{ij}^k$

$l := l + 1l := l + 1$

End for

End for

Output: Various tiers of clusters for each complementary mode $\{T_i^l\} \{T_i^l\}$.

3. Empirical results**3.1. Data source**

We collected three categories of data for each subregion in Zhejiang Province in 2023: (1) natural resources, including the area of land, water, forest and maritime space (as shown in Table 2); (2) renewable energy resources, including horizontal solar radiation and 10-m height annual average wind speed; and (3) installed capacity of renewable energy power stations, including their operational scale and specific geographic locations. The subregions in this study refer to county-level administrative units, which constitute a fundamental tier of China's administrative division system. Zhejiang Province consists 11 prefecture-level cities, and 90 county-level divisions, which correspond exactly to the subregions defined herein. Our dataset covers the natural resource endowments and renewable energy resource potential for all 90 county-level subregions.

Data on water areas (including rivers, lakes, and reservoirs) for each subregion in 2023, were derived from the LandSat8 dataset available via the ArcGIS platform. Collected by the Landsat 8 satellite, this dataset represents the longest temporal and most comprehensive open-access

Table 2

Abbreviations for different types of resources.

Resource	Abbreviation	Definition
Land Resource	Dryland area	Area of Dryland
	ALML area	Area of Arable land with medium to low yields
	Mountain area	Area of mountains with an altitude of >500 m
	MAML area	Area of mountain agricultural land with medium to low yields
Water Resource	IW area	Area of inland water
	TF area	Area of tidal flat area
Forest Resource	GMH area	Area of grassland with medium to high coverage
	Shrubland area	Area of Shrubland
	AAWS	Average annual wind speed
Maritime Space	Maritime area	Area of maritime space

Landsat 8 data archive covering China. Data on other land types and the average annual wind speed at 10 m above ground level for each subregion in 2023 were obtained from the Resource and Environment Science and Data Centre of the Chinese Academy of Sciences. In addition, annual total solar radiation data for each subregion were retrieved from the SOLARGIS database. Specifically, we adopted the long-term climatological average product (derived from satellite observations spanning 2007-2023), which characterizes typical solar resource conditions while mitigating the impacts of interannual variability. With a spatial resolution of approximately 4 km, solar radiation data were assigned a 2023 timestamp for consistency with other datasets in this study.

3.2. Data analysis**3.2.1. Descriptive analysis**

We conducted a preliminary statistical analysis of the collected data by comparing statistics such as the means, medians, variances, and other measures. Table 3 presents the distribution of natural resources in Zhejiang Province. The mean area of ALML is the greatest. The greatest regional variation was observed in IW and MAML. In general, there is substantial variation in the types of land across regions, which highlights the need to develop different complementary mode strategies for PV while considering land resources.

3.2.2. Visual analysis using heatmaps

To facilitate data visual analysis, we created heatmaps for the natural resources using ArcMap (ArcGIS 10.7) and inverse distance weighting by extrapolating cell values. ArcMap 10.7 is a core application in the ArcGIS Desktop suite released by Esri in March 2019. It is widely used for cartography, editing, analysis, and data analysis.

The heatmaps of different natural resource distributions in Zhejiang Province are shown in Figs. 3a and 3b by color. For instance, Figs. 3a indicates that the area of Dryland is intensive in northwestern Zhejiang Province. More specifically, drylands are intensive in regions such as Dinghai, Jiangshan, Yongjia, Chun'an, and Lanxi. Grassland with medium and high coverage, along with inland water, shows similar distribution patterns, all of which are intensive in northwestern Zhejiang Province (Figs. 3b & Figs. 3c). The shrublands are located primarily among the mountainous and hilly areas in the western and central parts of Zhejiang Province and show a decreasing trend from west to east (Figs. 3e). The total area of ALML in Zhuji and Lin'an is greater than 800 km² (Figs. 3b). In general, the wind resources in Zhejiang Province are mostly distributed along the coast.

3.3. Clustering results

The sum of squared errors (SSE) for the different PV complementary modes, as determined using the elbow approach, are shown in Fig. 4. The SSE decreases with K and eventually reaches a stable state. The decrease in the SSE becomes less significant as K increases, and the trend

Table 3

Distribution of natural resources.

Index	Mean	MSE	CV	Min	Max
Drylands area (km ²)	36.12	3.13	82.24	0	110.76
ALML area (km ²)	315.2	22	66.1	0	991.4
Mountains area (km ²)	255.9	45.3	167.81	0	2062.5
MAML area (km ²)	13.43	3.55	250.71	0	237.32
IW area (km ²)	0.02	0.01	543.19	0	1.28
TF area (km ²)	5.23	1.21	219.12	0	86.92
GMH area (km ²)	24.9	2.73	103.96	0	144.35
Shrublands area (km ²)	15.83	2.35	140.76	0	161.52
AAWS (km/h)	7.57	0.34	42.75	4	22.5
Maritime area (km ²)	0.03	0.15	543.19	0	1.28
Solar Radiation (KWh/m ²)	1319.60	43.10	3.26	1022.8	1417.9

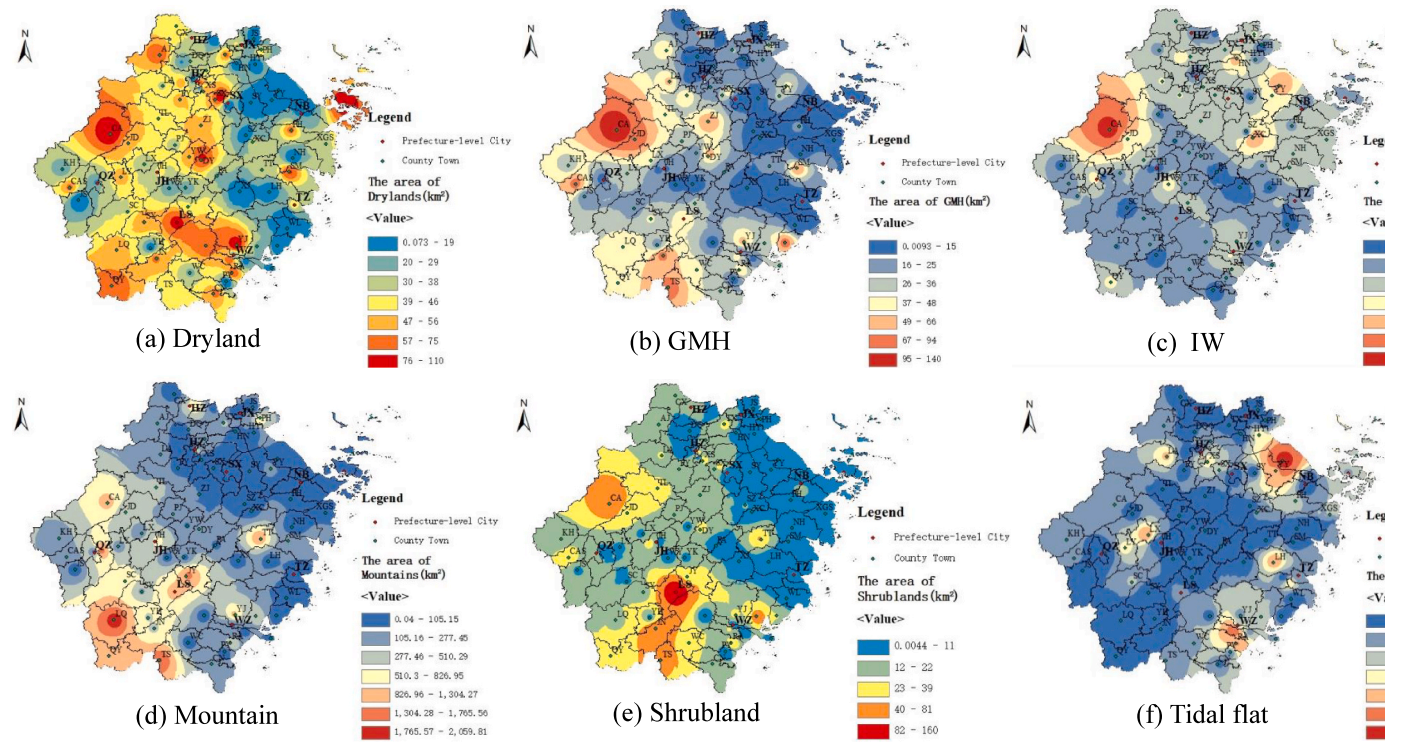


Fig. 3a. Heatmaps of natural resources in Zhejiang Province.

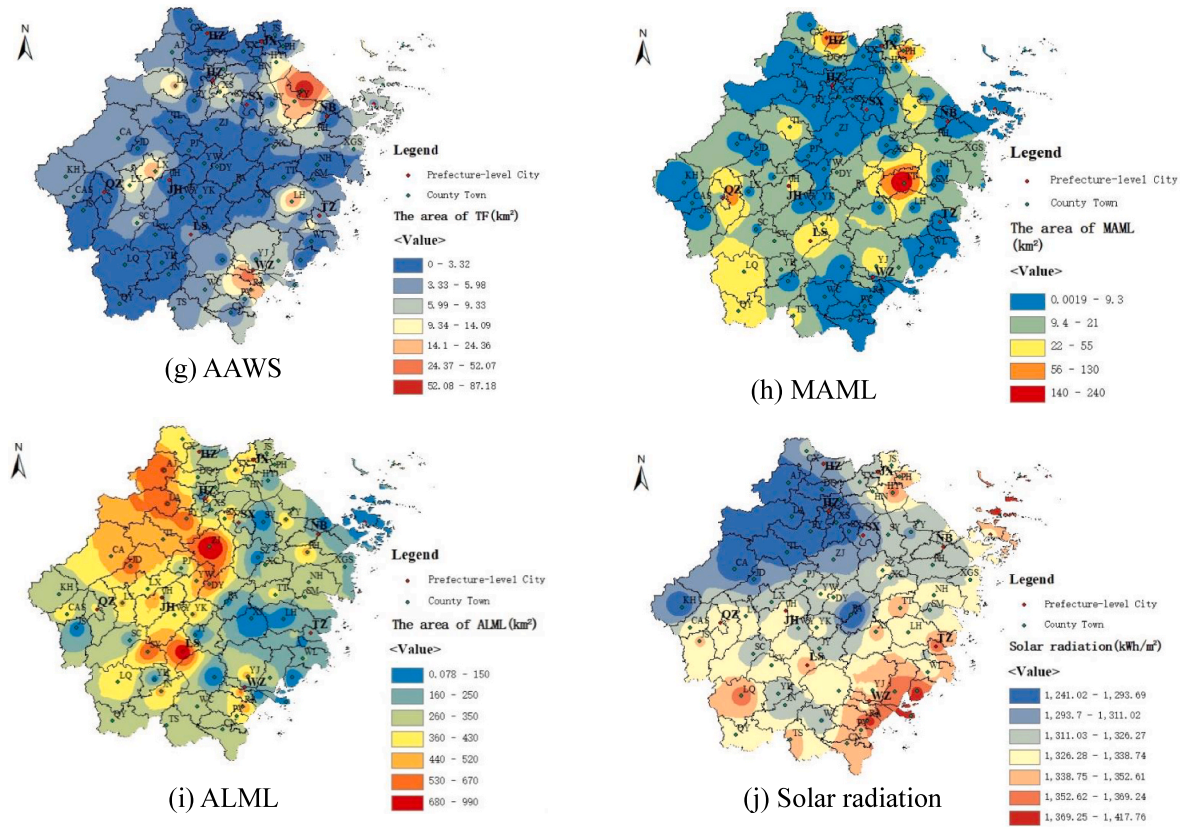


Fig. 3b. Heatmaps of natural resources in Zhejiang Province.

becomes less pronounced. In other words, as demonstrated in Fig. 4, the SSE decreases drastically as K increases from 11 to s , whereas it decreases gradually when K exceeds $s + 1$. Therefore, the ideal value for K

should be determined as the optimal number of clusters, denoted as s , by identifying the elbow point of the plot. Table 4 summarizes the ideal value of K for each complementary mode.

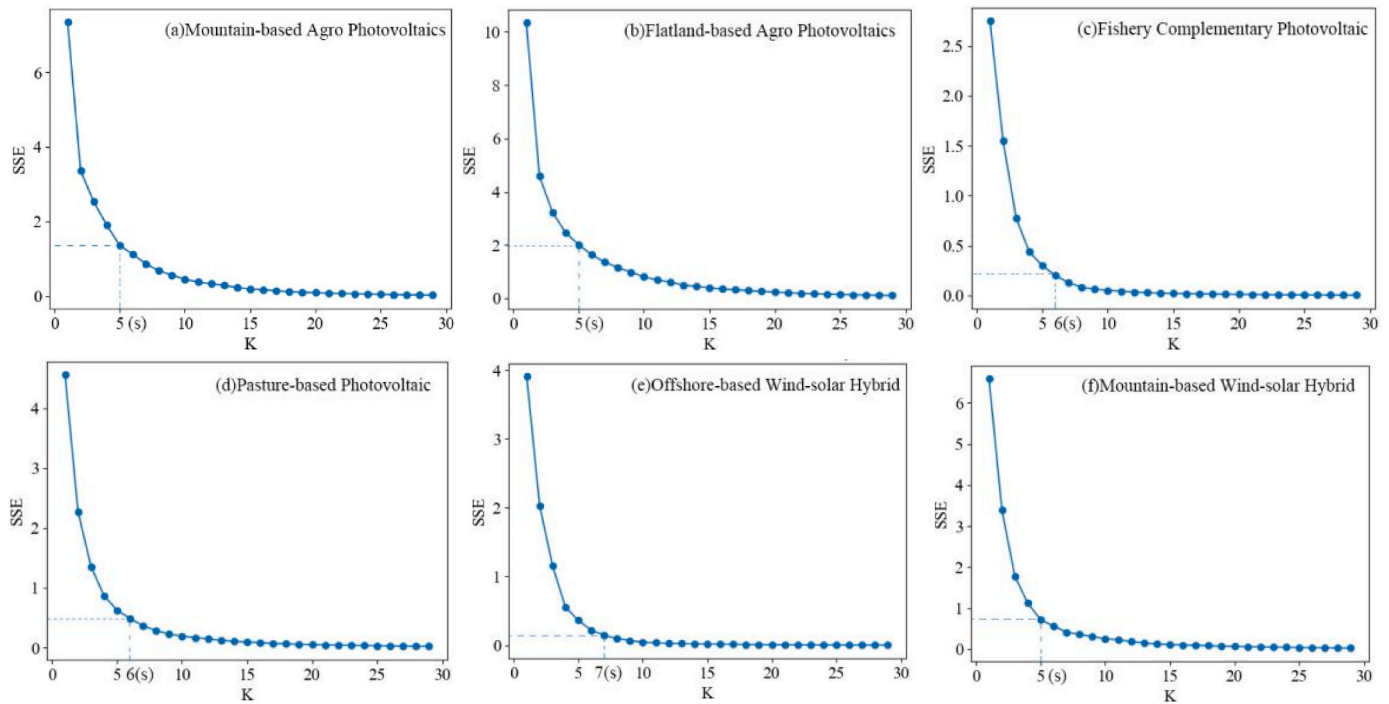


Fig. 4. Plots of the SSE under different complementary modes.

Table 4

Optimal values of K for different PV complementary modes.

Complementary Mode	Features of clustering	K
MAPV	The area of mountains, the area of MAML and the area of shrublands	5
FAPV	The area of drylands and the area of ALML	5
FPV	The area of TF and the area of IW	6
PPV	The area of GMH and the area of shrublands	6
OVS	AAWS and the area of maritime space	7
MWS	AAWS and the area of mountains	5

After determining the value of K for each complementary mode, we performed a cluster analysis to group subregions according to indices characterizing their features. As shown in Fig. 5, under a certain mode, there are points of different colours, with each point representing a subregion and each colour representing a cluster.

The differences in features among clusters with respect to the three complementary modes are represented as follows.

(1) Agrophotovoltaics

In accordance with three features, namely, the area of mountains, the area of MAML, and the area of shrublands [24], each subregion was assigned to its own cluster for MAPV. Mountainous areas with elevated terrains and moist climates provide favourable conditions for the cultivation of a variety of crops, such as mushrooms [25]. The clustering results are shown in Fig. 5(a), which reveals that the $C_1^4C_1^0$ category has the highest values across all three features, whereas the $C_1^0C_1^0$ category has the lowest values. In addition, the FAPV mode is appropriate for subregions with a significant expansion of ALML or Dryland, as stated by Junedi et al. [26]. Consequently, we performed clustering analysis on two features, namely, the area of drylands and the area of ALML, to determine the most appropriate category, represented as the $C_2^4C_2^0$ category in Zhejiang Province (Fig. 5(b)). In most subregions, these two features are at medium to low levels. Table A1 in Appendix A provides a comprehensive list of the subregions that are categorized into a unified

cluster for each complementary mode.

(2) Photovoltaic for breeding

FPVs are particularly suitable for subregions with large water areas or extensive tidal flats [27]. We conducted a clustering analysis on two features: the area of the TF and the area of the IW. As illustrated in Fig. 5 (c), the C_3^5 category has significantly elevated values in both dimensions, whereas the C_3^0 , C_3^1 and C_3^2 categories have nearly identical values. Moreover, we conducted a clustering analysis on two additional features: the area of the GMH and the area of shrublands [28]. Subregions with extensive areas of grassland and shrubland are conducive to grazing [29]. These results are shown in Fig. 5(d). The C_4^4 and C_4^5 categories exhibit the highest values for both features. Table A1 presents the subregions that are included in each category.

(3) Wind-solar hybrid

We conducted a clustering analysis on two features: AAWS and the maritime area [30]. The data in Fig. 5(e) illustrate that AAWS exhibits a high degree of uniformity across various subregions. The C_5^6 category is equipped with massive AAWS and a large maritime area, whereas the C_5^0 category ranks poorly in both dimensions. In addition, we conducted a clustering study using two other features: AAWS and the area of mountains [31]. In terms of AAWS, the C_6^6 category has notably higher values, whereas the C_6^5 and C_6^0 categories have the worst performance in the dimension of the area of mountains (Fig. 5(f)). Table A1 presents the subregions encompassed within each category for various complementing modes.

3.4. Significance tests for differences between classes

On the basis of the results of the clustering analysis presented in Fig. 5, each subregion was assigned to a specific cluster. Subsequently, Mann-Whitney U tests were conducted to determine whether the differences between clusters were statistically significant. The results of these significance tests are detailed in Table 5 and the tables in Appendix

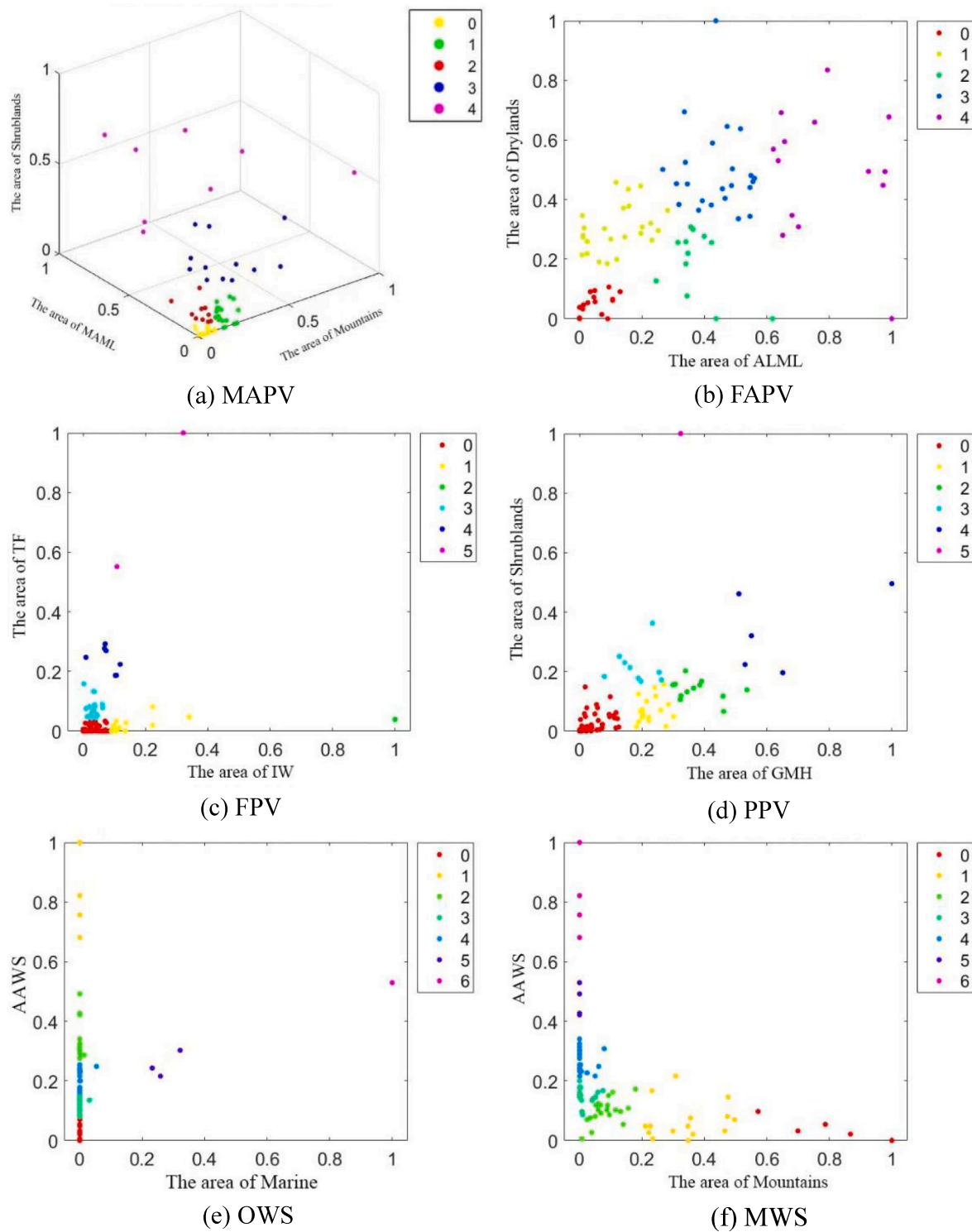


Fig. 5. Results of K-means clustering.

B.

(1) **MAPV:** There is a statistically significant difference at the 5% level between the cluster group {1, 3, 4} and the other clusters in terms of the area of shrublands. With respect to the areas of ALML and drylands, Cluster 4 stands out from the other four clusters at a significance level of 5%. As shown in Table 5, the comprehensive ranking was determined by summing and sorting the tier orders for each category across the three MAPV indices. Cluster 4 ranks

highest, with a total of 3 (1 + 1+1), whereas Cluster 0 ranks lowest, with a total of 11 (3 + 4+4). Thus, the subregions in Cluster 4 are the most favourable for the development of the MAPV mode.

(2) **FAPV:** Clusters 3 and 4 exhibit similar areas of ALML, both of which are ranked in the first tier. However, there is a notable 5% disparity in terms of the area of drylands between them, with Cluster 4 ranked at the first tier. As shown in Table 5, the comprehensive ranking was determined by summing and sorting

Table 5

Cluster prioritization according to the significance tests.

Complementary modes	Index (area)	First tier	Second tier	Third tier	Fourth tier	The prioritized Clusters
MAPV	Shrublands	1, 3, 4	2	0		4
	Mountains	4	3	1, 2	0	
	ALML	4	2, 3	1	0	
FAPV	Drylands	4	2, 3	1	0	4
	ALML	3, 4	1	2	0	
FPV	TF	5	4	3	0, 1, 2	5
	IW	2	1, 5	4	0, 3	
PPV	Shrublands	5	3, 4	2	0, 1	4, 5
	GMH	4	2, 5	1, 3	0	
OWS	Maritime	5, 6	0, 1, 2, 3, 4			1, 6
	AAWS	1	2, 6	4, 5	0, 3	
MWS	Mountains	0	1	2	3, 4, 5, 6	/
	AAWS	6	5	4	0, 1, 2, 3	

the tier orders for each cluster across the two FAPV indices. Cluster 4 ranks highest, with a total of 2 (1 + 1), whereas Cluster 0 ranks lowest, with a total of 8 (4 + 4). Thus, the subregions in Cluster 4 are the most favourable for the development of the FAPV mode.

- (3) **FPV:** Cluster 5 ranks at the first tier with respect to the area of TF, while it ranks at the second tier with respect to the IW area. As shown in Table 5, Cluster 5 ranks highest, with a comprehensive ranking of 3 (1 + 2), whereas Category 0 ranks lowest, with a total of 8 (4 + 4). Thus, the subregions in Cluster 5 are the most favourable for the development of the FPV mode.
- (4) **PPV:** Cluster 5 ranks at the first tier with respect to the area of shrublands, whereas Cluster 4 ranks at the first tier regarding the GMH area. As shown in Table 5, both Clusters 4 and 5 rank highest, with a comprehensive ranking of 3 (1 + 2), whereas Category 0 ranks lowest, with a total of 8 (4 + 4). Thus, the subregions in Clusters 4 and 5 are the most favourable for the development of the OWS mode.
- (5) **OWS:** Clusters 5 and 6 rank in the first tier with respect to the marine area, whereas Cluster 1 ranks in the first tier with respect to the AAWS area. As shown in Table 5, both Clusters 1 and 6 rank highest, with a comprehensive ranking of 3 (1 + 2), whereas Categories 0 and 3 rank lowest, with a total of 6 (2 + 4). Thus, the subregions in Clusters 1 and 6 are the most favourable for the development of the OWS mode.
- (6) **MWS:** Implementation of the MWS mode in Zhejiang Province is challenging because of the lack of subregions with abundant wind energy resources and extensive mountainous regions. As shown in Table 5, the Mann–Whitney *U* test results indicate that none of the clusters are ranked in the top tiers across all the indices. Overall, no subregion is appropriate for the development of the MWS mode.

3.5. Potential of subregions for complementary development

3.5.1. Classification of potential subregions

On the basis of the results of clustering and statistical analysis, we further identify the potential of subregions for different types of complementary modes and elaborate on their implementation.

Category I Subregions with the potential for developing MAPVs. Compared with the other groups, a notably greater proportion of low-yield arable land and dryland facilitates the efficiency of land utilization. These subregions, which are primarily located in the central and northern regions of Zhejiang Province, have well-established economies and significant prospects for the extensive implementation of the MAPV.

Category II Subregions with the potential for developing FAPVs. Most of these subregions are located in the hilly region of southwestern Zhejiang, where high-elevation and humid environments create ideal growing conditions for a wide variety of mushrooms [32]. Thus, these regions are capable of implementing a three-dimensional PV land use strategy that involves the generation of electricity on PV panels and the cultivation of Chinese herbals beneath these panels.

Category III Subregions with the potential for developing FPV. Within the administrative boundaries of these subregions, a large area of inland water exists. The integration of PV electricity generation and fishery farming involves installing PV panels above the water surface to generate electricity and cultivate special aquaculture below the water surface. This provides benefits such as solar shading, reducing the evaporation of water bodies and improving the agricultural environment to minimize the economic and environmental expenses associated with the complementary system.

Category IV Subregions with the potential for developing PPV. These subregions are equipped with extensive grassland areas with medium to high coverage. Thus, deliberately placing PV panels above the breeding farm a viable option, which can be implemented locally. Moreover, this measure can contribute to maximizing the utility of local lands for both breeding and generating PV electricity while meet growing energy and food demands efficiently.

Category V Subregions with the potential for developing OWSs. These subregions possess numerous offshore wind resources. By capitalizing on the abundance of available space to install wind farms, one can leverage the pristine condition of the ocean, which includes extended periods of sunlight, optimal utilization of light energy and a distinctive environmental benefit that has the potential to increase power generation. In addition, the distance to regions with high energy demand is short, thus contributing to the low financial burden of electricity transmission.

Category VI Subregions with the potential for developing MWS. The significance test and clustering results indicate a lack of subregions in Zhejiang Province that exhibit excellent potential for developing MWS, mostly owing to the discrepancy between the wind speed and mountainous areas of the regions.

3.5.2. Prioritization hierarchy of subregions and economic efficiency

On the basis of the specified analysis, we provide the most favourable PV complementary modes for each subregion through visualization. We focus on the first and second tiers, as shown in Fig. 6, because the third and subsequent tiers are of limited development potential. The data depicted in the figure strongly correlate with the resource distribution

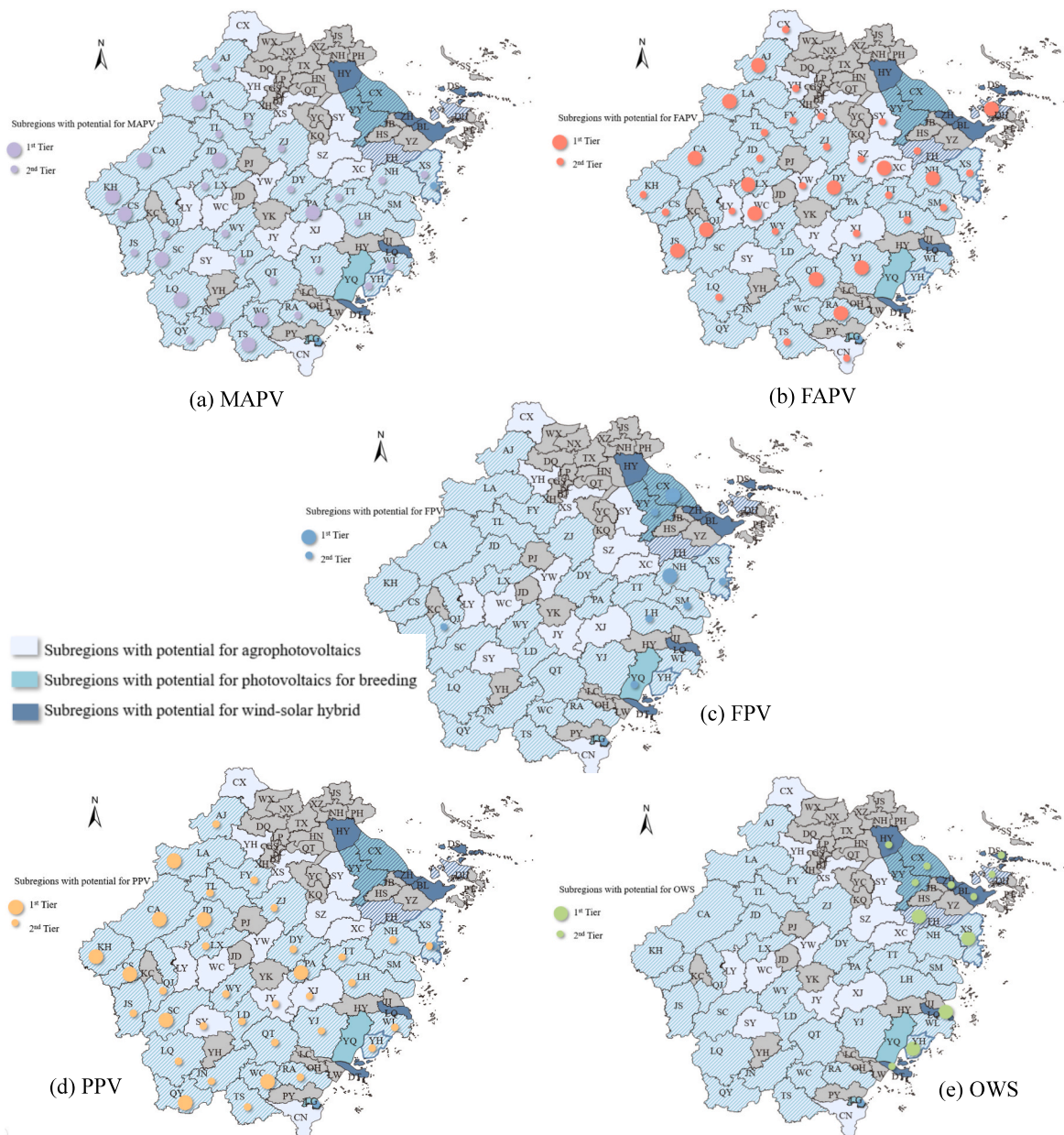


Fig. 6. Potential of subregions for complementary development in Zhejiang Province (Note: Grey areas represent subregions that are unsuitable for the aforementioned modes.).

heatmap (see Figs. 3a and Figs. 3b). The northwest region of Zhejiang Province (first-tier cluster) possesses significant amounts of dryland, GMH, and ALML resources, making it suitable for the development of APV. Moreover, the northeast region of Zhejiang Province (second-tier cluster), specifically Cixi and Yuyao, possesses a plentiful supply of TF and IW resources, making it suitable for developing FPVs. Therefore, the use of data analysis and visualization can help local governments accurately assess the potential for PV development in different subregions and provide effective guidance for government policy formulation.

Moreover, prioritization hierarchy is closely related to the insights from Fig. 2. Under a complementary mode, subregions ranked in the 1st tier often exhibit a high degree of consistency in their complementary resource advantages. For instance, the subregion LQ shows high potential in the MAPV mode because its indicators for mountainous area and shrubland both fall within high-value clusters in the clustering analysis (as shown in Figs. 3a and Figs. 3b). Moreover, there is also a

counterintuitive finding: under a certain mode, if a region possesses a distinct advantage in one natural resource, it may still exhibit high development potential even without advantages in other resources. For instance, KH exhibits high potential for developing MAPV complementary mode. This occurs because the region demonstrates significant disadvantages in mountain and shrubland resources but holds a clear advantage in ALML resources.

At present, some subregions in Zhejiang Province have successfully integrated PV energy generation with local industry, in accordance with the findings presented in Fig. 6. For instance, Zhoushan city, which is recognized for its potential for FPV, has successfully implemented the integration of fishery and PV. This integration enables both to progress in tandem and ensures a stable energy supply during peak seasons. Annual power generation is expected to reach 126 million kWh, leading to savings of approximately 43,000 tonnes of standard coal and a reduction of 106,000 tonnes of carbon dioxide emissions per year. Moreover, the city of Cixi has initiated the development of PV facilities

for 1.8 billion Chinese yuan (¥). These facilities are spread throughout a water area of 4492 acres and have a total installed capacity of 200 MW. The PV panels in Pingdu city are installed above the crops, enabling optimal land utilization. The PV power station has an average yearly electricity generation of 180 MWh, generating a revenue of 71 million Chinese yuan (¥), and is expected to mitigate carbon dioxide emissions by 140 million tonnes per year. This study introduces methodologies to facilitate the empirical identification of a novel complementary mode for the advancement of renewable energy. Therefore, the proposed DCD framework, which has the potential to simultaneously achieve both environmentally friendly electricity generation and agricultural yield, may promote renewable energy economics. Moreover, it is worth mentioning that the clustering outcomes, including the optimal number of clusters and subregion groups, determine the structure of the subregion cluster tiers.

4. Discussions

4.1. Marginal contributions

In our study, using data-driven methodology, we constructed a map of the distribution of potential for the development of PV in complementary modes throughout 90 subregions in Zhejiang Province (as shown in Fig. 6). The results indicate that most subregions in Zhejiang Province are favourable for adopting at least one type of complementary mode to facilitate the growth of the PV industry. There are substantial marginal contributions to the field of knowledge in the following three areas.

- (1) From the perspective of land resource utilization, most current studies have focused on the implementation and effects of colocating PV systems alongside industrial activities on the same land parcel. For example, Gasch et al. [33] developed an evaluation index system for PV agriculture and introduced concepts such as agrivoltaics (APV) and PV-assisted breeding. Other researchers have examined key drivers and challenges in APV implementation [34], as well as economic and technological factors associated with installing PV systems in vineyard trellis interrows [35]. Additional studies have investigated the effects of PV-based complementary industrial modes, including the environmental effects of PV systems in fisheries [19], digitalized business models and the economic benefits of floating photovoltaics (FPVs) [36], and the influence of PV installations on livestock in pastoral settings [37]. In contrast, this study proposes a DCD framework to evaluate the potential of complementary PV modes at the subregional level in land-constrained areas such as Zhejiang Province, China.
- (2) From the perspective of complementary photovoltaic (PV) development modes, research on enhancing the performance of multienergy complementary systems (MECSs) in the context of complementary photovoltaic (PV) development has expanded in recent years. For instance, Nascimento et al. [17] evaluated the theoretical and technological potential, as well as the complementarity, of offshore wind-solar (OWS) energy in Brazil. Studies such as that by Ren et al. [38] have adopted complementary modes such as wind-PV, hydro-PV, hydro-wind, and hydro-wind-PV in MECSs. Moreover, Li et al. [5,22,39] conducted a detailed assessment of the spatiotemporal distribution and characteristics of resources associated with OWS energy. In this study, we examine the available energy resources in land-constrained regions, such as Zhejiang Province, China, and investigate six complementary modes, namely, wind energy, solar energy, fishery, agriculture, and animal husbandry.
- (3) From the perspective of potential identification methodology, traditional research has focused mainly on applying K-means clustering to group entities into clusters [8]. However, we

proposed a DCD framework that achieves statistically more robust potential identification through the systematic integration of K-means clustering, the Mann–Whitney *U* test, and the proposed STS algorithm.

4.2. Policy recommendations

According to the results illustrated in Fig. 6, we can draw the following implications for policy recommendations.

Implication 1 Developing the PV complementary modes in a gradual and orderly manner considering long-term planning and tailored to specific subregions and cluster tiers.

A key issue in developing complementary PV industries is the widespread neglect of synergies between photovoltaic systems and region-specific sectors. During the expansion of the PV sector, inadequate attention has been given to fostering other complementary industries, such as agriculture. It is essential that the development of PV complementary systems does not impede the growth of existing industries. To address these challenges, it is recommended that governments formulate long-term plans informed by local natural resource endowments and market demand. This should involve strategic planning of the PV industry layout, identifying priority development modes for different cluster tiers or subregions, scaling up projects in advantageous areas, and piloting initiatives in less developed regions to assess their potential. Subregions with the highest potential for a specific complementary mode (first tier in Fig. 6) should be encouraged to pioneer its implementation, thereby generating valuable insights for subsequent followers (second tier in Fig. 6). However, not all subregions are suitable for the development of the six complementary modes. These grey areas should not encourage the complementary development of the PV industry.

Our research also offers the following implications for the Zhejiang Provincial Government's policy design.

Implication 2

- 1) Subregions (category I) located in the central and northern regions of Zhejiang Province with large proportion of low-yield arable land and dryland have high potential for developing MAPV.
- 2) Subregions located in hilly region of southwestern Zhejiang (category II) is feasible to implementing a three-dimensional PV land use strategy that involves the generation of electricity on PV panels and the cultivation of Chinese herbals beneath these panels.
- 3) Subregions with a large area of water (category III) is suggested to develop the integration of PV electricity generation and fishery farming.
- 4) Subregions with extensive grassland areas with medium to high coverage (category IV) are potential for developing PPV.
- 5) Subregions possess numerous offshore wind resources (category V) are potential for developing OWSs.
- 6) No subregion is potential for developing MWS.

5. Conclusion

It is challenging for regions with land resource constraints to promote PV power generation because of their inherently high land requirements. Therefore, a green–green dilemma arises, highlighting a contradiction between conserving land resources and achieving the development goal of green energy. To address this challenge, the industry usually resorts to seeking alternative solutions, including fostering complementary relationships between PV and other industries (such as agriculture and animal husbandry) that share the same land.

In this study, on the basis of the characteristics of land resources in Zhejiang Province, China, we summarized six complementary modes appropriate for developing the PV industry. These six complementary modes are MAPV, FAPV, FPV, PPV, OWS, and MWS. To identify the potential of each subregion of Zhejiang Province for the development of each complementary mode, we proposed a strategy formulation framework, namely, DCD, to promote the development of solar energy by effectively using the limited natural resources available. The DCD framework involves incorporating renewable energy production facilities, namely, PV in this study, with existing primary industries to enhance their interdependent development. This framework uses the K-means algorithm to group 90 subregions into clusters for each complementary mode, and the Mann–Whitney *U* test for significance verifies the difference between two clusters and the developed STS algorithm to determine the potential of subregion clusters for the development of each complementary mode.

Taking Zhejiang Province in China as an example, we collected data involving 3 types of resources from 90 subregions: (1) natural resources in terms of 10 indices, (2) renewable energy resources in Zhejiang Province, and (3) the installation of renewable energy stations. To apply the DCD framework, we selected the appropriate indices and grouped the 90 subregions into clusters for each complementary mode. For instance, concerning the MAPV mode, three indices (namely, the area of mountains, the area of MAML, and the area of shrublands) were used to group the subregions into five clusters. Afterwards, considering all three indices, we employed the proposed STS algorithm to rank the potential of clusters in terms of cluster tiers for the development of each complementary mode. The first tier exhibited the highest potential, whereas the fifth tier had the lowest potential. The results of the DCD framework provide implications for the local governments of the subregions and even Zhejiang Province in terms of determining industry development policies and progress. The observation of some real-world projects partially verifies the results of the potential analysis of Zhejiang Province.

We contribute to the literature and the practice of renewable energy by proposing six PV-based complementary modes for the development of the PV industry by considering the endowment of natural resources. These modes can also provide valuable insights for regions constrained by natural resources. Moreover, by integrating the K-means clustering algorithm with the STS algorithm, we proposed a data-driven complementary-based strategy framework for the development of the PV industry within regions facing (land) resource constraints. Without loss of generality, this framework is applicable to the development of other types of renewable energy facing the constraints of natural resources. Finally, the application of the DCD framework suggests that Zhejiang Province is well suited for prioritizing five complementary modes for developing the PV industry, particularly APV and PV for breeding.

There are several limitations that merit attention in future work. First, while our analysis focuses on PV-based complementary modes, existing research has not yet considered all possible complementary modes, which warrants further investigation. Second, while this study concentrates on natural resource constraints, future work could be extended to incorporate human-made environments—such as cities with rooftop, parking, facade, and roadside installations—as well as floating PV systems on reservoirs [40]. Third, our study did not consider specific local information. Future research could also expand the assessment of the development potential of PV complementary modes from the perspective of local information, including the level of economic development, the financial status of the government, detailed land topography, specific aquaculture species, local community support, and grid connection availability, and the willingness of farmers. Fourth, the unavailability of data from other similar scenarios precluded us from conducting a more extensive validation of the generalizability of the DCD framework. However, given its theoretical rationality, we are confident that the framework holds promise for application across a broader range of scenarios. Fifth, there may be potential biases in the

data, such as biases in the remote sensing data. Therefore, addressing concerns related to the accuracy and resolution of the data can be considered a critical future direction. Finally, the data on wind speed and solar radiation are subject to limitations in terms of temporal scope. In this study, we used data from 2022, which may have fluctuated across different years. However, this does not affect the applicability of the DCD framework. A promising direction for future research would be to extend temporal analysis from a dynamic perspective.

CRediT authorship contribution statement

Yun Zhou: Conceptualization, Funding acquisition, Writing – original draft, Methodology, Project administration, Investigation. **Xinran Yin:** Data curation, Writing – original draft, Project administration, Investigation. **Yifan Chen:** Writing – original draft, Validation. **Jian Cao:** Conceptualization, Funding acquisition, Writing – original draft, Investigation. **Enjing Jiang:** Data curation, Writing – original draft. **Yuxiu Si:** Visualization, Writing – original draft. **Xuanlan Huang:** Data curation, Formal Analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence this work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.esr.2026.102128>.

Data availability

Data will be made available on request.

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